

setting up a semiconductor fab in India are significantly higher than other potential locations like China and Taiwan. "This is due to the lack of a robust electronic components and semiconductor manufacturing ecosystem in India, which leads to dependence on imports for raw materials, equipment, and tools. Moreover, the cost of power, water, and transportation is high in India. Additionally, the cost of land acquisition and environmental clearance can delay the project implementation and increase capital expenditure," explains Sanjay Gupta, Chairman of the India Electronic and Semiconductor Association (IESA).

Industry veteran and VLSI Society president Satya Gupta, who visited the US recently to meet leaders from across the value chain, says, "Political stability, and more importantly, policy stability and longevity in case there is a political change, was one of the most important doubts in people's minds. The political, bureaucratic, and technical leadership have to create a better awareness of India's resolve to create a long-term sustainable and commercially viable semiconductor ecosystem in India and provide assurances that political leadership changes will not affect the policies and incentives."

Another factor is the availability of raw materials. Access to electronic components is a big challenge. "We are heavily dependent on imports for raw materials, which adds to transport costs, inventory management, and logistics," says Sanjeev Keskar, Executive Director at EPIC (Electronics Products Innovation Consortium) and former president of IESA.

SCALING DEMAND

Countries like the US, Europe, and Japan enjoy one other advantage: the significant presence of fabless



"Semiconductor manufacturing is not a today or tomorrow game. It is a day after tomorrow and down the road game... decades-long game"

RAJEEV CHANDRASEKHAR
MINISTER OF STATE, MINISTRY OF ELECTRONICS AND IT



"Two types of companies can help India's chip mission—global established chip companies and Indian conglomerates"

SATYA GUPTA
PRESIDENT, VLSI SOCIETY

and electronics product companies, which are the customers of chip makers. This makes it easier for chip manufacturers to expand operations and build new fabs.

Given the current situation, Satya Gupta adds, there are two types of companies that can help India's Semiconductor Mission. First there are established companies like Micron, Intel, etc. Second, and more importantly, large Indian conglomerates.

"Recently, there have been some indications about Reliance Industries exploring investments in this area. A company like Reliance or Tata [group] with the deep experience of executing large projects, global reach, long investment mindset, financial muscle power, and their focus on digital technologies and customers will be a very potent recipe."

One source of hope for India is the emergence of an electronics manufacturing ecosystem over the past decade. Playing to its strengths and taking advantage of countries' desire to diversify their supply chain beyond China, India has managed to lure some of the biggest names—like Apple, Samsung, HP, Dell, and Lenovo—to set up assembly lines in India.

Taking a leaf out of that playbook perhaps, India has also been aggressive in pitching the country to semiconductor majors. From PM Modi to Ministers of MeitY Ashwini Vaishnaw and Chandrasekhar, they have all held multiple consultations with stakeholders to apprise them of the developments on the policy front. Besides, in the past two years, the government has addressed gaps in infrastructure. Sources at MeitY tell *Business Today* the government is evaluating some more proposals.

India will hope that these come through, and soon. **BT**

@nidhisingal